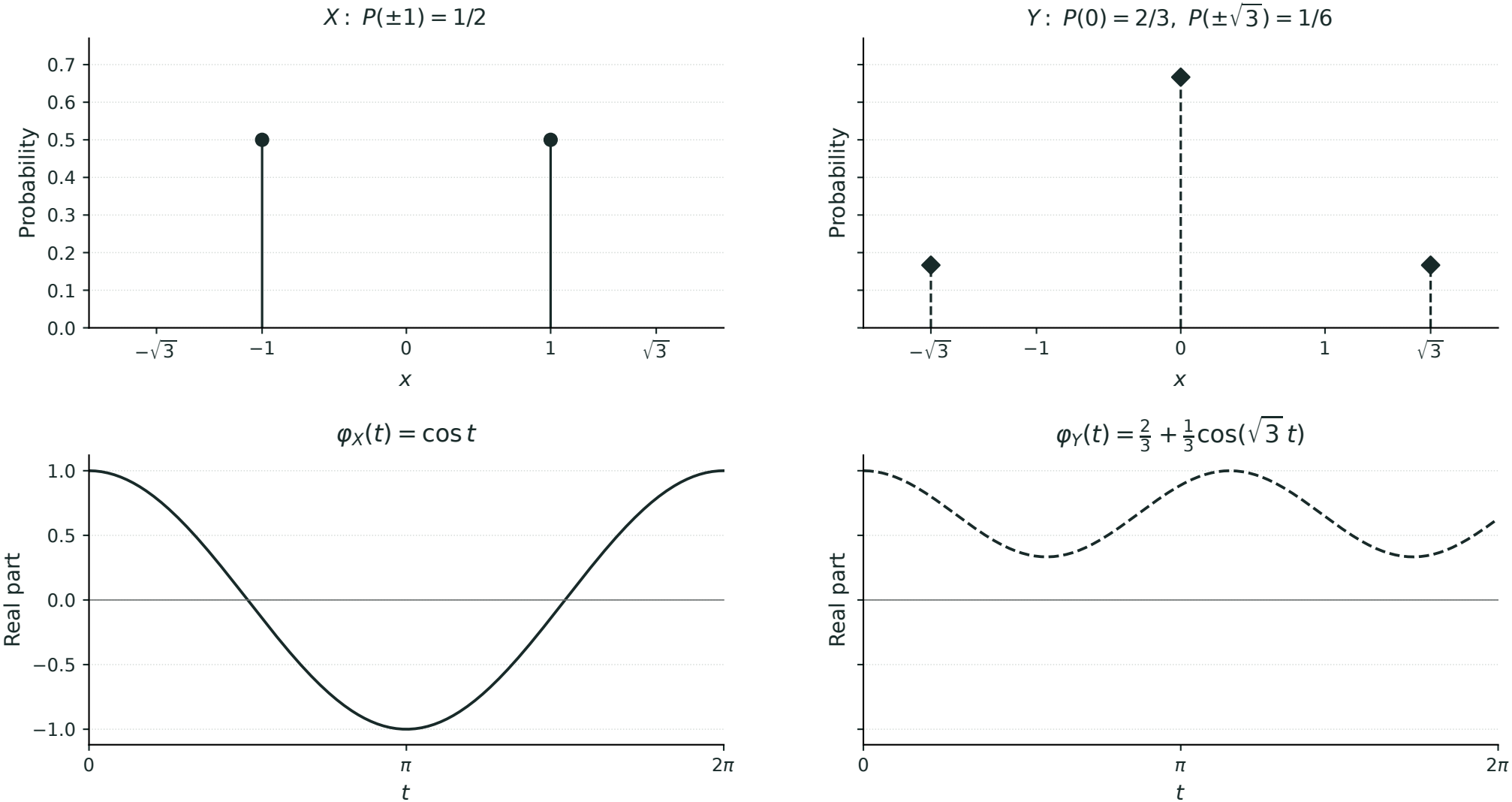
The same mean and variance do not identify a distribution



$\mathbb{E}X = \mathbb{E}Y = 0$, $\mathrm{Var}(X) = \mathrm{Var}(Y) = 1$. The full characteristic function identifies the law; these two functions differ.